

Roberto Eisenlohr

☀️ AI Solutions Architect | Quant Researcher & Developer | Enterprise Financial Systems

☎️ +1 (347) 688-8396

✉️ admin@abfs.tech

🌐 www.abfs.tech

📄 linkedin.com/in/robertoeisenlohr

📁 Portfolio

📍 Manhattan, New York, NY



Summary

AI Solutions Architect with 23+ years of experience as a professional Engineer. Building production systems at the intersection of **artificial intelligence** and **financial markets**. Recently completed an 11-month engagement at **Citigroup** where I led development of an enterprise GenAI Ecosystem for portfolio analysis — serving C-level operations across equities, options, currencies, and futures.

Core Expertise:

- **Agentic AI Systems** — LangGraph cyclic reasoning, multi-model orchestration (Gemini, Claude, Grok, Ollama), MCP server integration
- **Financial Domain** — Portfolio analysis, risk assessment, regulatory compliance, algorithmic trading
- **Full-Stack Delivery** — Architecture → Development → Deployment → Monitoring (Python, AWS, Docker, FastAPI)

Expert in building autonomous agentic systems for risk assessment, portfolio analysis, and regulatory risk review using cutting-edge Gen-AI integration, multi-agent frameworks, and large language models. Proven capability to integrate and leverage advanced technologies (LLMs, MCP Servers, AWS, GCP Vertex, Gemini, Claude, Ollama) to deliver robust, real-time solutions for quantitative teams. Skilled at dynamic Python code generation, debugging, and execution, automated HTML/CSS reporting, and end-to-end deployment of cloud-based platforms.

Credentials: Two Engineering degrees, MBA coursework in Finance & Corporate Law (FGV, IBMEC), Harvard AI Certification, SmythOS AI Agent Engineer, FINRA Series 57, 32 Professional Certifications



Experience



Citigroup / Citibank

Mar 2025 – Jan 2026

Quantitative Developer – GenAI Architect

- Architected enterprise GenAI Ecosystem for portfolio analysis serving C-level operations across the institution.
- Designed and implemented agentic AI frameworks using Python and LangGraph, enabling autonomous reasoning and acting loops for complex financial workflows.
- Integrated multiple MCP (Model Context Protocol) servers for seamless tool orchestration across trading systems and data sources.
- Built intelligent agents for risk assessment, portfolio analysis, and regulatory risk review using multi-model architectures (Gemini, Claude).
- Developed, implemented, and supported large-scale platforms and agentic systems that operate across multiple departments, countries, companies, and portfolios, tailored for C-Level operations and decision-making.

- Program autonomous agents to interpret Quant queries, dynamically write, debug, and execute Python code in real time, and iteratively enhance solutions through runtime analysis.
- Develop agents capable of automatically generating and refining HTML/CSS visualizations, facilitating clearer interpretation of quantitative solutions for quants.
- Led end-to-end systems development: from ideation and initial design through programming, implementation, and deployment—including LLM model training and integration.
- Extensive hands-on experience with SQL, AWS (S3, EC2), Google Vertex AI, Gemini, Claude, building complex AI-powered platforms and agentic systems.
- Managed financial assets: Equity Options, Stocks, Credit, ETFs, Currencies, and Futures.
- Led a team of 7 high-performance Engineers throughout building the Ecosystem.
- **Saved quantitative research teams thousands of hours annually through intelligent automation of research workflows.**
- **Reduced time-to-insight for quantitative research from days to minutes through intelligent task decomposition and parallel AI agent execution via MCP protocol.**
- Collaborated with Quant Developers, Quant Analysts, Directors, Risk Analysts, Portfolio Managers, and Quant Traders, delivering transformative solutions to multi-disciplinary teams.



ABFS TECH

March 2016 – Present

Quant Researcher, Developer & AI Architect

- Developed and implemented AI-driven trading systems outperforming the S&P 500 by 3X over 10 years.
- Engineered a grid system with ROI averaging 10% monthly, Sharpe Ratio of 4.7.
- Designed and implemented AI trading systems for hedge funds, brokerage firms, family offices, proprietary traders.
- API integrations for Charles Schwab, TD Ameritrade, Interactive Brokers, IG, FOREX.com, Pepperstone.
- Developed in Python, C#, C++, NinjaScript 8, MQL5, MQL4, ThinkScript, Pine Script.
- Programmed GenAI applications—Google Gemini, Anthropic's Claude, Opus, xAI's Grok, OpenAI's ChatGPT, Ollama.
- Integrated financial platforms, built smartphone applications, connected systems within financial markets.
- Extensive expertise with AWS cloud computing solutions.
- Worked closely with Portfolio Managers, Front Office, Middle Office, Back Office, and teams from multiple Hedge Funds, Family Offices, Banks, and other financial institutions.
- Automated anomaly detection and root cause analysis using GenAI.
- **Provided quant research, development, and technology services to multiple clients across the financial markets, collaborating and delivering solutions tailored for each client's unique requirements, diverse asset classes, and business strategies.**



T3 Trading Group

Feb 2019 – Dec 2019

Quant Researcher & Developer (Part-Time)

- Developed algorithmic trading solutions for Stocks, Equity Options, ETFs using machine learning/statistics.
- Built, backtested, optimized trading algorithms and financial performance tests in Python, SQL, C#, C++.
- Led API integrations for real-time trading and data workflows.
- Partnered with Portfolio Managers, Traders, Developers to deliver production strategies.

Freelance Software Developer | Fully Remote

July 2015 – February 2016

Full-Stack Developer & Technical Consultant

- Delivered custom software solutions for clients across financial services, e-commerce, and technology sectors.
- Built web applications and APIs using Python, C#, JavaScript, SQL with focus on scalable architecture and clean code.
- Developed trading tools and portfolio analytics dashboards for individual traders and small investment firms.
- Implemented database designs, RESTful APIs, and front-end interfaces for data visualization.

- Managed full project lifecycle from client requirements gathering through deployment and maintenance.

Brinpa Investments and Participations
Investments Director (Part-Time)

Jan 2006 – Jun 2015

- Managed early-stage investments, following up through final payout.
- Responsible for performance analysis and investment project selection.

AEP Engineering
Lead Engineer

Jan 2002 – Dec 2014

- Led a team of Engineers that designed, implemented, and maintained ISO 9000 and other Quality Assurance certifications with regular external audits, as a requirement for the Engineering firm to work with a bank.
- Built tools and dashboards to visualize and communicate data quality, performance monitoring, and insights.
- Strong knowledge in statistics, probabilities, statistical analysis, and probabilistic analysis.

 Key Achievements

Agentic AI Framework & MCP Infrastructure: Planned, developed, and deployed enterprise agentic system with Model Context Protocol (MCP) architecture, building custom MCP servers and clients that orchestrate multiple AI operations and saved quantitative teams thousands of hours annually by automating research workflows	Execution Optimization: Reduced latency by 95% and improved implementation shortfall by 15% for equity options flow
Risk Model Deployment: Built ML-based risk models reducing portfolio drawdown by 18% while maintaining target returns	Quantitative Research: Developed systematic trading strategy with Sharpe Ratio of 4.7 and 10% average monthly returns over multi-year live trading period
3X S&P 500 Outperformance: AI-driven trading systems outperforming benchmark over 10-year backtest with rigorous walk-forward validation	Real-Time Risk Monitoring: Implemented production risk system alerting on portfolio breaches within 100ms of market data updates

 Skills

Programming Languages Python C# C++ SQL HTML CSS JavaScript MQL5 MQL4 NinjaScript 8 ThinkScript Pine Script	Software & Tools Amazon Web Services (AWS) Google Gemini Anthropic's Claude Opus Sonnet OpenAI's ChatGPT Ollama Metatrader Charles Schwab Trading Platform TradingView Alveo C# Platform Unity
Quantitative Analysis	Trading Strategies

Algorithmic Trading

Quantitative Research

Mathematical Modeling

Statistical Analysis

Risk Management

Portfolio Optimization

Backtesting

Derivatives Pricing

Monte Carlo Simulation

Time Series Analysis

Financial Algorithms

Options & Futures

ETFs & Indexes

Pearson Correlation Analysis

Statistical Arbitrage

Pairs Trading

Mean Reversion

Momentum

Market Making

Execution Algorithms

Grid Trading

Trend Following

target achievement

Triangular Arbitrage

Dynamic Circular Arbitrage

300+ trading strategies developed

Risk Management

VaR (Value at Risk)

CVaR (Conditional Value at Risk)

Greeks Analysis (Delta, Gamma, Theta, Vega)

Scenario Analysis

Stress Testing

Drawdown Control

Position Sizing

Artificial Intelligence (AI)

Machine Learning

AI Trading Systems

Predictive Modeling

Neural Networks

Model Optimization

Deep Learning

Natural Language Processing

Large Language Models (LLM)

Generative AI (GenAI)

Agentic AI

LangGraph

LangChain

LangSmith

Model Context Protocol (MCP)

Financial Expertise

Financial Engineering

Quantitative Finance

Financial Modeling

Cryptocurrency Trading

Forex Trading

Equities

CFDs

Portfolio Management

Financial Analysis

Equity Options

Stocks

Futures

Futures Options

Credit Instruments

Currencies

Other Skills

API Integration

Mobile App Development

Project Management

Client Communication

Strong Analytical & Research Skills

Problem-solving

Excellent Written & Verbal Communication

Low Latency Development

Docker

FastAPI

Playwright

 Education

Harvard University

Certification, Artificial Intelligence in Python
August 2023 – January 2024

Federal Fluminense University (UFF)

1st Engineering Degree
Diploma, 2002

UNESA

2nd Engineering Degree
Diploma, 2014

Ibmec

Attended 50 credits towards MBA, Finance
2005–2006

FGV – Getulio Vargas Foundation

Attended 36 credits towards MBA, Corporate Law
2003–2004

 Certifications & Licenses

AI & Machine Learning Certifications

Certification / License	Issuing Organization	Date
Model Context Protocol: Advanced Topics	Anthropic	Jan 2026
Beyond Vibe Coding with Google Gemini: Become a Deliberate Developer	LinkedIn Learning	Dec 2025
AI Agent Engineer	SmythOS	Sep 2024
Harvard University Certification, Artificial Intelligence in Python	Harvard University	Jan 2024
Prompt Engineering for ChatGPT	Vanderbilt University	Feb 2024
Artificial Intelligence Certification	Google Launchpad	—
Machine Learning Certification	Google Launchpad	—
Data Science Certification	Google Launchpad	—

AWS Certifications (6)

API Serverless Gateway Certification	AWS Cloud Computing Certification	Decoupled Applications Certification
Deploying & Scaling Applications with Amazon Lightsail Certificate	Lambda Foundations Certification	MySQL Certificate

Google Launchpad Certifications (18)

Artificial Intelligence Certification	Blockchain Certification	Cybersecurity Certification	Data Science Certification
Docker Certification Course	Firebase Certification	GIT Certification	HoneyCode Certification
Machine Learning Certification	Node.js Certification	Programming in C# Certification	Python 3 Certificate
Python Advanced Certification	SQL Programming Certification	Social Media Marketing Certification Course	
Solidity Programming for Ethereum Blockchain Certification	Unity Programming Certificate		
Website Building Certification Course			

Other Certifications & Licenses

Licensed Engineer – Crea-RJ	PHP Programming Certification – Sololearn
Series 57 - Securities Trader Representative Exam License (previously held) – FINRA	WooCommerce Themes Certification

Total: 32 Professional Certifications

Publications

Cryptocurrency – Starter's Guide for Traders and Investors

ISBN: 978-1-63901-260-2

Publisher: Primedia

Publication Date: May 2021

Available at: <https://cryptocurrency-book.com/>